

Hemanth Balakrishna Nuthalapati

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SUMMARY

- Software Engineer with 3+ years of experience building secure and scalable backend and full-stack applications using Java (Spring Boot, Spring MVC), Node.js, Angular, and React.
- Strong background in designing RESTful APIs and microservices, with deep experience in Oracle PL/SQL and hands-on application of OWASP/CWE secure coding standards and DevSecOps practices.
- Skilled in setting up and automating CI/CD pipelines with Jenkins, GitHub Actions, Docker, Kubernetes, and Maven, and experienced in deploying applications on AWS (EC2, ECS, RDS, Lambda, S3) and Azure (AKS, Functions, Blob Storage).
- Proficient in database design, query optimization, and performance tuning across Oracle, PostgreSQL, MongoDB, and Redis, ensuring reliability in high-volume transactional environments.
- Experienced in system design, unit and integration testing (JUnit5, Mockito, Cypress, JMeter), and monitoring production systems with Splunk, Prometheus, and Dynatrace to maintain performance at scale.
- Strong collaborator in Agile/Scrum/SAFe environments, with proven experience in leading code reviews, mentoring teammates, and delivering end-to-end full-stack solutions aligned with business goals.

TECHNICAL SKILLS

Languages	Java, SQL, TypeScript, JavaScript (ES6+), Python
Backend & APIs	Spring Boot, Spring MVC, Spring Security, Node.js, REST, GraphQL, Microservices, API Gateway Design
Frontend	Angular (TypeScript, HTML5, CSS3), React (components, state management), JSP, Responsive UI/UX
Databases	Oracle (SQL/PL-SQL, Stored Procedures, Query Optimization), PostgreSQL, MongoDB, Redis
Testing & Quality	JUnit5, Mockito, Jasmine, Cypress, Postman, JMeter (Performance Testing), SonarQube, Checkmarx, OWASP Secure Coding, TDD/BDD Practices
Cloud & DevOps	AWS (EC2, RDS, ECS, Lambda, S3, IAM), Azure (Blob, AKS, Functions), Docker, Kubernetes, Jenkins, Maven, GitHub Actions, Terraform (IaC)
Messaging & Processing	Apache Kafka, RabbitMQ, Spring Batch
Monitoring & Logging	Splunk, Dynatrace, Prometheus, Kibana, OpenTelemetry (Distributed Tracing), Log4J
Other	OAuth 2.0, JWT, SAML, Gitflow, Agile/Scrum/SAFe, Design Patterns, System Design & Architecture

PROFESSIONAL EXPERIENCE

Software Engineer - *Big Eye Tech, LLC, Remote*

May 2025 – Present

- Designed and developed backend features and RESTful APIs using Spring Boot, Spring MVC, and Node.js, integrating with Oracle PL/SQL, PostgreSQL, and MongoDB. These services supported secure and reliable data flows that powered high-volume business transactions.
- Partnered with QA to validate APIs using Swagger and Postman, applying OWASP secure coding standards and enforcing SonarQube/Checkmarx checks. This proactive approach significantly reduced vulnerabilities before production.
- Delivered responsive and modular front-end components in Angular, React, and JSP, ensuring seamless integration with backend services and improving the overall user experience.
- Expanded the testing strategy with JUnit5, Mockito, Jasmine, Cypress, and integration tests, reducing regression issues by 25% and increasing confidence in production deployments.
- Tuned queries, built indexes, and redesigned schemas in Oracle and PostgreSQL, which reduced response times on high-traffic queries and allowed the system to scale as customer usage grew.
- Automated CI/CD pipelines on AWS and Azure with Docker, Kubernetes, Jenkins, GitHub Actions, and Maven; introduced Terraform templates and integrated Fortify/Checkmarx scans for stronger DevSecOps compliance.
- Actively led code reviews, design discussions, and mentoring sessions, fostering collaboration and promoting best practices across the engineering team.
- Integrated internal and third-party APIs to support financial and transactional workflows, ensuring compliance with industry regulations while maintaining high system availability.
- Partnered with monitoring and operations teams to improve observability by adding Splunk, Prometheus, Dynatrace, and OpenTelemetry, which accelerated incident resolution and root-cause analysis.

Software Engineer– *Changelab, Los Angeles, CA*

Sep 2024 – May 2025

- Built and optimized real-time dashboards and responsive front-end features using React, TypeScript, HTML5, and CSS3. These solutions were integrated with REST and GraphQL APIs to provide users with fast, interactive views of live business data.

- Designed and tuned backend services with Spring Boot, Spring MVC, and Node.js, improving database performance in Oracle PL/SQL, PostgreSQL, and MongoDB. By adding indexing and restructuring queries, I helped increase API throughput by 20% and reduce average response latency by 30%, which directly improved customer-facing performance.
- Implemented secure APIs with OAuth 2.0, JWT, and SAML, enabling seamless integration with external financial and identity management providers. This ensured compliance with enterprise security standards while supporting high transaction volumes.
- Strengthened reliability by expanding API testing with Postman, JUnit5, Mockito, Cypress, and JMeter. Combined with static analysis gates in SonarQube/Checkmarx and adherence to OWASP secure coding practices, this improved release quality and reduced production defects.
- Contributed to the automation of CI/CD pipelines on AWS and Azure, using Docker, Kubernetes, Jenkins, Maven, and GitHub Actions. To enforce security and compliance, I embedded Fortify and Twistlock scans and introduced Terraform templates to manage infrastructure as code.
- Applied well-known design patterns (Factory, Adapter, Strategy, Singleton) to ensure backend services were modular, scalable, and easy to maintain as requirements evolved.
- Worked in Agile/Scrum/SAFe teams, taking an active role in peer code reviews, sprint planning, and design discussions. I made sure technical designs aligned with business requirements and could scale securely as usage grew.

Software Engineer – Mphasis, India.

Jan 2020- Jun 2022

- Integrated front-end applications with backend APIs built in REST, GraphQL, and SOAP to support low-latency, high-performance data flows for large-scale financial systems that demanded high reliability and consistent uptime.
- Enhanced existing Spring Boot and Node.js microservices by tuning PostgreSQL and MongoDB queries. By adding indexes and rewriting stored procedures, I helped reduce execution times and improved throughput during peak workloads.
- Contributed to Spring Batch workflows that processed large volumes of financial data in near real time. Leveraged Kafka and RabbitMQ to handle event-driven processing, ensuring transactional pipelines scaled efficiently without data loss.
- Supported CI/CD automation by setting up deployments on AWS ECS and Azure AKS using Jenkins and Maven. Enforced SonarQube static analysis checks to maintain code quality and reduce production issues.
- Applied secure coding practices (OWASP, OAuth 2.0, JWT, SAML) and worked on production incident resolution. Collaborated with monitoring teams to analyze logs and metrics in Splunk, Log4J, and Prometheus, which helped identify root causes faster and reduce downtime.
- Worked closely with Agile/Scrum teams to deliver maintainable backend components. Applied design patterns such as Factory, Adapter, Observer, and Strategy to ensure the solutions were scalable and easy to extend as business needs evolved.

ACADEMIC PROJECTS

AI-Powered Document Classification System – Natural Language Processing & Deep Learning

- Developed an NLP-based classification model using Python, Scikit-learn, and TensorFlow to automatically categorize large document sets.
- Implemented BERT and Word2Vec embeddings for semantic understanding, improving classification accuracy by 20% over baseline models.
- Built a Flask-based API to integrate classification results into web applications, enabling real-time document tagging and search.

Stock Price Prediction System – Time Series Forecasting & Data Analytics

- Built a stock market prediction system using Python, Pandas, NumPy, and machine learning models (ARIMA, LSTM) with TensorFlow/Keras to forecast price trends.
- Preprocessed and normalized data, applied feature engineering, and achieved improved prediction accuracy through hyperparameter tuning.
- Developed interactive visualizations with Matplotlib, Seaborn, and Plotly for trend analysis and investment insights.

EDUCATION

CALIFORNIA STATE UNIVERSITY, LONG BEACH, CA

Master Of Science in Computer Science

Aug 2022 – May 2024

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY UNIVERSITY, CHENNAI, TN, INDIA.

Bachelor Of Technology in Computer Science

Aug 2017 – May 2021